

High-moment inertia, killed

A fixed even cycle-moment does not certify $\text{ind}_-(R_{N-3} - \frac{1}{2}I) \leq 1$

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Abstract

The dual resolvent contraction R_{N-3} of a canonical window is a positive contraction. For every integer $d \geq 1$ the pointwise bound $1_{\{r < 1/2\}} \leq [2(1-r)]^{2d}$ yields $\text{ind}_-(R - \frac{1}{2}I) \leq 2^{2d} \text{tr}((I - R)^{2d})$. If the right-hand side were < 2 for one *fixed* $d \in \{2, 3, 4, 6, 8\}$ on MAIN, Twin and living χ , with reserve growing in N , then P1 would be a finite source cycle-sum, outside the two-moment ceiling class. It is not. On the named census the moment is ≥ 16 already at $d = 8$ on FRAME-A, grows with N (core-42: M_2 from 34.9 at $N = 142$ to 224.5 at $N = 878$), and no d — even $d > 8$ — brings it under 2 on the $n_- = 1$ branch. The cause is a macroscopic cluster of eigenvalues of R_{N-3} at $\frac{1}{2}$ from above (FRAME-A: 36 in $[\frac{1}{2}, \frac{3}{5}]$), each contributing ~ 1 to the sum. Scramble sees $\text{ind}_- = 21$; two-period dies; dead χ is ≥ 2 as a control, but so is living χ — the test is not a world separator. No RH claim.

Status. Lemma 1 and Lemma 2 are proved (finite identities). Proposition 1 names the kill. No RH claim. No L^* claim. No $R^\dagger$ claim.

1 The object

Let R_{N-3} be the dual Christoffel–Darboux Gram of the first $N - 3$ dual orthogonal polynomials, restricted to the hole nodes Y — the rest block of the r367 two-rank cut [1, 2]. On a window (μ, ν) with half-filling $S = 2N - 1$,

$$R_{N-3} = B_{\bullet, 0:N-3} B_{\bullet, 0:N-3}^T,$$

B the dual OP matrix dressed by $\sqrt{u^\vee}$ and evaluated on Y . Haynsworth reduces window-local $R^\dagger \succ \frac{1}{2}I$ to (P1) $\text{ind}_-(R_{N-3} - \frac{1}{2}I) \leq 1$ and an $O(1)$ edge sign [3]. The present test is a *sufficient* majorant for (P1), preregistered before any source-Weyl energy estimate.

Hard stops, binding: d is chosen from the five-element set $\{2, 3, 4, 6, 8\}$, never a function of N ; the certificate is a trace of an even power, not an eigenvector; no target margin enters.

2 The bound

Lemma 1 (even moment majorant). *Let $0 \preceq R \preceq I$ on a finite-dimensional real Hilbert space, with eigenvalues r . For every integer $d \geq 1$ and every $r \in [0, 1]$,*

$$1_{\{r < 1/2\}} \leq [2(1-r)]^{2d}.$$

Summing, $\text{ind}_-(R - \frac{1}{2}I) \leq 2^{2d} \text{tr}((I - R)^{2d})$.

If $r < \frac{1}{2}$ then $2(1-r) > 1$, so the even power is ≥ 1 . If $r \geq \frac{1}{2}$ the indicator vanishes and the right-hand side is nonnegative. Over $\mathbb{Q}$: $r = \frac{2}{5}$, $d = 2$ gives $[2(1 - \frac{2}{5})]^4 = (\frac{6}{5})^4 = \frac{1296}{625} \geq 1$. Dropping the prefactor 2^{2d} makes the bound false: $(1 - \frac{2}{5})^4 = \frac{81}{625} < 1$. An odd power is not a majorant without the contraction hypothesis: $R = \text{diag}(\frac{2}{5}, 2)$ has $\text{ind}_- = 1$ but $\sum_i [2(1 - r_i)]^5 < 0$.

3 Cycle sums

Lemma 2 (closed walks). *For a real symmetric A ,*

$$\mathrm{tr}(A^k) = \sum_{i_1, \dots, i_k} A_{i_1 i_2} \cdots A_{i_k i_1},$$

the sum of weighted closed walks of length k on the complete graph with edge weights A_{ij} . On the dual CD graph, $A = I - R_{N-3}$ and $k = 2d$ even.

Exact on a 2×2 toy: $A = \begin{pmatrix} 1/2 & 1/4 \\ 1/4 & 1/3 \end{pmatrix}$, $\mathrm{tr}(A^2) = \frac{35}{72}$. On a 3×3 numerical toy the $k = 4$ walk-sum matches $\mathrm{tr}(A^4)$ to 10^{-17} . Because the kill-test fails, this identity is *not* the R399 target: a source expansion of $\mathrm{tr}((I - R)^{2d})$ would bound a quantity that is not < 2 .

4 The census, KILL_FAIL

Proposition 1 (KILL_FAIL). *On every named living window, for every $d \in \{2, 3, 4, 6, 8\}$, $M_d := 2^{2d} \mathrm{tr}((I - R_{N-3})^{2d}) \geq 2$. The reserve $2 - M_d$ shrinks as N grows. No registered d therefore reduces P1 to a cycle-moment theorem.*

FRAME-A ($w = 9$, $N = 184$, $|Y| = 104$): $0 \preceq R_{N-3} \preceq I$ (gate), $\mathrm{ind}_- = 1$, $r_{\min} = 0.46153$,

$$M_2, M_3, M_4, M_6, M_8 = 33.48, 26.79, 22.75, 18.23, 16.03.$$

Eigenvalues in $[\frac{1}{2} - \delta, \frac{1}{2} + \delta]$: 5, 24, 37 for $\delta = 0.01, 0.05, 0.1$; 36 lie in $[\frac{1}{2}, \frac{3}{5}]$. The r367 rest $A_0 + \frac{1}{2}I$ agrees with R_{N-3} at residual 0.

Core-42: 42/42 contractions, $\mathrm{ind}_- \in \{0, 1\}$, $M_2 \in [33.22, 224.5]$. At smallest $N = 142$ ($kz = 18$), $M_2 = 34.86$; at largest $N = 878$ ($kz = 52$), $M_2 = 224.5$ — reserve shrinks. The MAIN-85 sample continues the growth (kz119: $M_2 = 287.9$; EXT $kz = 69$, $N = 5690$: $M_2 = 1079$). Twin $kz \in \{9, 18, 52\}$: $|M_2^{\mathrm{twin}} - M_2^{\mathrm{MAIN}}| \leq 2 \cdot 10^{-5}$, same inertia.

Living χ_3 (37 windows) $M_2 \in [28.03, 183.4]$; living χ_4 (41 windows) $M_2 \in [26.90, 178.7]$; all ≥ 2 . Dead sprouts $\chi_3 \in \{15, 19, 23, 33, 39\}$ and $\chi_4 = 20$ are likewise ≥ 2 (the control holds) — but living worlds fail too, so the test is not sharp as a world discriminator.

5 Why it dies, and why no larger d saves it

Write $M_d = \sum_i [2(1 - r_i)]^{2d}$. Modes with $r > \frac{1}{2}$ have $2(1 - r) < 1$ and decay in d ; a mode with $r < \frac{1}{2}$ has $2(1 - r) > 1$ and *grows* in d . On FRAME-A the minimum over $d \in \mathbb{N}$ is $M_{11} = 15.06$, still $\gg 2$, then $M_{30} = 90$. On $kz = 18$ ($r_{\min} = 0.390$) the minimum is $M_5 = 28.66$, then $M_{30} = 1.5 \cdot 10^5$. On the vacuous branch $\mathrm{ind}_- = 0$ ($kz = 12$, $r_{\min} = 0.500018$) the bound is monotone decreasing, but the cluster sits at distance $O(10^{-5})$ from $\frac{1}{2}$: even the bulk-only sum drops below 2 only at $d \sim 1024$, which grows with $1/\mathrm{gap}$ and is forbidden.

The two-moment ceiling class cannot see this cluster; a growing even moment cannot either, once d is not allowed to track N . P1 remains an integer mode count. The next object is source Weyl energy of the centred mass difference after the three edge modes, as preregistered, not a cycle-sum of $I - R_{N-3}$.

6 Named kills

Scramble (matched χ -comb, seed 1): $\mathrm{ind}_- = 21$, $r_{\min} = 0.0036$, $M_2 = 148$, $M_8 = 2.65 \cdot 10^5$. The trace sees the overload; M_d explodes in d . Two-period ($S = 21$, $c = \frac{2}{3}$): $\mathrm{ind}_- = 4$, $r_{\min} = 0$, $M_2 = 52$. Mutants: 2^{2d} omitted fails over $\mathbb{Q}$; odd exponent fails on a non-contraction. Dead χ is ≥ 2 , as required of a sharp test — and uninformative here, because living worlds are ≥ 2 as well.

Machine check

Standalone: Fractions bound, nopref, odd mutant, 2×2 cycle-sum, frozen d -set, 3×3 walk-sum. Construction: FRAME-A contraction and r367 residual, moments, scramble 21, two-period, dead χ_{3-15} , living χ_{3-9} with $\text{ind}_- = 0$. Discovery census: `high_moment_inertia.probe.py` (`--smoke / full`), core-42, MAIN sample, Twin, living and dead χ . Exit line: HIGH MOMENT VERIFIED.

Claim boundary

Finite identities for the even-moment majorant and for closed walks; a named census that the sufficient test of P1 fails on every registered d , with reserve shrinking in N ; named scramble / two-period / mutant / dead- χ controls. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

- [1] S. Hamann, *Final two-rank inertia*, sealed probe, August 2026, `experiments/tfpt-discovery/final_two_rank_inertia.probe.py`.
- [2] S. Hamann, *Mixed Haynsworth*, sealed probe, August 2026, `experiments/tfpt-discovery/mixed_haynsworth.probe.py`.
- [3] S. Hamann, *The saturated dual resolvent*, standalone note, August 2026, `rh/problem/rdagger_saturation.tex`.